VaR Analysis Report: Indian Paint Sector Portfolio

1. Executive summary

The workbook estimates one-day Value at Risk (VaR) for a hypothetical ₹10,000,000 equity portfolio of seven Indian paint-sector stocks, using three methods (parametric, historical simulation, Monte Carlo). It then backtests each method and applies the Basel traffic-light test, and summarises everything on a dashboard.

Headline results as the workbook reports them:

- 95% one-day VaR sits between ₹198,000 and ₹216,000 (about 2.0% to 2.2% of portfolio value).
- 99% one-day VaR sits between ₹287,000 and ₹314,000 (about 2.9% to 3.1%).
- Historical simulation is the best-calibrated method in the backtest. Parametric and Monte Carlo both under-predict 99% breaches (20 observed against about 12 expected).
- All three methods land in the Basel Yellow zone (5 to 6 breaches in the last 252 days).

Most important finding from my independent review: the daily returns in the workbook have their sign reversed. The data is sorted newest-first, but the log-return formula divides the older price by the newer one, so every return is the negative of the true return (and is labelled one trading day early). I recomputed everything from the price data and the workbook's figures reproduced exactly with the same method, which confirms the cause. Correcting it raises the historical 99% VaR from ₹314,289 to ₹356,631 (about 12% higher) and changes the backtesting picture. Details are in section 7.

2. Portfolio and data

The portfolio is weighted by market capitalisation (total ₹2.9175 trillion). Weights are fixed and entered once on the Data sheet.

Stock	Market cap (₹ trn)	Weight	Daily volatility
Asian Paints	2.0805	71.31%	1.42%
Berger Paints	0.4923	16.87%	1.53%
Indigo Paints	0.0366	1.25%	1.86%
Kansai Nerolac	0.1444	4.95%	1.62%
Shalimar Paints	0.0035	0.12%	2.73%
Dulux (Akzo Nobel India)	0.1362	4.67%	1.68%
Sirca Paints	0.0240	0.82%	2.48%

- **Sample:** 1,240 daily closing prices from 1 April 2021 to 1 April 2026, giving 1,239 daily log returns.
- **Portfolio value:** ₹10,000,000 (hypothetical).
- **Portfolio statistics (workbook):** mean daily return 0.0168%, daily standard deviation 1.240% (about 19.7% annualised).

3. How the workbook is built

Sheet	Purpose
Data	Prices, market caps and market-cap weights
Returns	Daily log returns per stock, =LN(P_t / P_previous row)
Portfolio Returns	Weighted daily portfolio return via SUMPRODUCT
Parametric VAR	Normal-distribution VaR: mean + z × standard deviation, with z = -1.645 (95%) and -2.326 (99%)
Historical VAR	Empirical 5th and 1st percentiles of portfolio returns
Monte Carlo Stimulation	1,000 simulated returns using NORM.S.INV(RAND()), mean and volatility taken from the parametric sheet
Model Comparison	Side-by-side VaR table with a bar chart
Backtesting	Daily breach flags for each model and confidence level, with expected against actual breach counts
Basel Traffic Light Test	Last 252 observations at 99%, classified Green (0-4), Yellow (5-9) or Red (10+)
Dashboard	Portfolio summary, VaR metrics, Basel table and two charts

Everything is formula-driven, and each VaR sheet produces a plain-English interpretation sentence, which makes the workbook easy to follow.

4. VaR results as reported in the workbook

Model	95% VaR (return)	95% VaR (₹)	99% VaR (return)	99% VaR (₹)
Parametric	-2.023%	202,307	-2.868%	286,822
Historical simulation	-1.984%	198,397	-3.143%	314,289
Monte Carlo	-2.156%	215,593	-2.919%	291,931

Reading these: the three methods agree closely at 95% and spread out at 99%, where tail shape matters most. Historical is highest at 99%, which hints that the real distribution has a heavier tail than the normal assumption behind the other two. The Monte Carlo figures are random and will change every time the workbook recalculates (see section 7).

5. Backtesting (1,239 days)

Expected breaches are 61.95 at 95% and 12.39 at 99%.

Model	95% actual	95% difference	99% actual	99% difference
Parametric	59	-2.95	20	+7.61
Historical	62	+0.05	13	+0.61
Monte Carlo	47	-14.95	20	+7.61

A Kupiec proportion-of-failures test on these counts (my calculation) gives:

- **Historical:** passes at both levels (p = 0.99 at 95%, p = 0.86 at 99%).
- **Parametric:** passes at 95% (p = 0.70) but is rejected at 99% (p = 0.046).
- **Monte Carlo:** rejected at both levels (95%: p = 0.042 for being too conservative; 99%: p = 0.046). Part of this is simulation noise from using only 1,000 draws.

The Historical model's near-perfect score is partly mechanical. It is tested on the same data its percentiles were estimated from, so it will match by construction (see section 7).

6. Basel traffic-light test (last 252 days, 99%)

Model	Breaches	Zone
Parametric	6	Yellow
Historical	5	Yellow
Monte Carlo	6	Yellow

All three sit in the Yellow zone (5 to 9 breaches), where a regulator would typically ask for review and may apply a higher capital multiplier. The Zone column on the Dashboard (F10:F12) is currently empty, so the classification is not shown there.

7. Review findings

7.1 Sign error in the return calculation (high priority)

The Returns sheet lists dates newest-first, and the formula =LN(B3/B2) divides the price in the current row (older) by the price in the row above (newer). That produces ln(P_old / P_new), the negative of the true return. For example, the value shown against 30 March 2026 is -2.76% for Asian Paints, but the price actually rose from ₹2,165.20 (30 March) to ₹2,225.80 (1 April).

I rebuilt the series from the Data sheet in both ways. The workbook's method reproduced its mean, standard deviation and both percentiles exactly, and the corrected series is its mirror image: mean -0.0168% instead of +0.0168%, skew -0.33 instead of +0.33.

Why it matters: the left tail is the whole point of VaR, and the workbook is currently reading the right tail of true returns.

Measure	Workbook	Corrected	Change
Parametric 95% VaR	₹202,307	₹205,664	+₹3,357
Parametric 99% VaR	₹286,822	₹290,179	+₹3,357
Historical 95% VaR	₹198,397	₹200,726	+₹2,329
Historical 99% VaR	₹314,289	₹356,631	+₹42,342 (+12%)
Parametric 99% breaches (1,239 days)	20	26	+6
Basel breaches, Parametric	6 (Yellow)	5 (Yellow)	
Basel breaches, Historical	5 (Yellow)	1 (Green)	

The parametric shift is exactly twice the daily mean applied to the portfolio value. The large change is in the historical 99% figure, because the true left tail is fatter than the mirrored right tail.

With the corrected series, the parametric 99% model fails the Kupiec test much more clearly (26 breaches against about 12 expected, p = 0.0007). Historical still passes. I did not recompute Monte Carlo because it is random, but it uses the same mean and volatility as the parametric model, so it behaves similarly.

The fix is to reverse the ratio so it reads =LN(B2/B3) in a newest-first layout (or sort oldest-first and keep the current formula), then re-align the dates.

7.2 The normal distribution does not fit

On the corrected series, skewness is -0.33 and excess kurtosis is 3.60, and a Jarque-Bera test rejects normality overwhelmingly. The worst day was -7.49% (25 May 2022) and the best +6.31% (14 May 2021). Normal-based VaR therefore understates the tail, as the 99% parametric backtest shows. The workbook does not report Expected Shortfall. On the corrected historical series, ES is about ₹302,700 at 95% and ₹463,800 at 99%, well above the VaR figures.

7.3 Concentration limits diversification

Asian Paints and Berger together are 88% of the portfolio and contribute roughly 95% of portfolio variance (Asian Paints about 80%, Berger about 15%). The other five stocks add about 5% of the risk combined. Correlation between Asian Paints and Berger is 0.57; most other pairs are 0.14 to 0.30. In effect, this is a two-stock risk profile with five small additions, which is worth stating in the interpretation.

7.4 Monte Carlo design

- It draws a single standard normal per day and scales it by the portfolio's mean and volatility. It does not simulate the seven stocks with their correlations, so it carries no information beyond the parametric model, apart from sampling noise.
- RAND() is volatile, so the 1,000 draws change on every recalculation. The Model Comparison, Backtesting, Basel and Dashboard figures drift with them, while the written interpretations (for example the Backtesting comments) are typed text that will not update.
- 1,000 draws is small for a 1% quantile (only about 10 tail observations).

7.5 Backtests are in-sample

The VaR thresholds are estimated on the full five years and then tested on those same days, including the 252-day Basel window. That favours the historical method and gives an optimistic read. A proper test estimates VaR on data prior to each test day (rolling or expanding window).

7.6 Data and housekeeping

- Several price cells on the Data and Returns sheets are text with thousand separators (for example '2,225.80'). They work through Excel's implicit conversion but are fragile.
- Prices may be unadjusted. Some single-day moves are very large (Sirca +18.2% on 4 Oct 2021; Shalimar 16.4% on 1 Dec 2023; Nerolac 16.4% on 2 Aug 2022; Dulux about -14.6% on 17 Dec 2025). Worth checking each against corporate actions (splits, bonuses, special dividends, listings).

- Hardcoded values: 1239 (observation count in Backtesting and Dashboard), 7 (number of stocks) and the data-period label. Link these to the data.
- Labels: the sheet is named "Monte Carlo Stimulation" (should be Simulation) and a cell reads "Portofolio".
- Weights are static market-cap weights from one date, applied across five years, and returns are weighted log returns, which is a small approximation to simple-return weighting.

8. What the workbook does well

- Clear sheet-by-sheet flow from prices to dashboard, with live formulas throughout.
- Three standard VaR methods compared at two confidence levels.
- Backtesting with expected against actual breach counts, and the Basel traffic-light framework applied.
- Plain-English interpretation lines tied to the computed values.
- A dashboard that puts the headline numbers and charts on one page.

9. Recommended next steps

1. **Fix the return sign** and re-run all sheets; update the dashboard and comments with the corrected numbers.
2. **Fill the Dashboard Zone column** and make the backtesting interpretations formula-driven.
3. **Replace the Monte Carlo** with a correlated multi-asset simulation (Cholesky decomposition of the covariance matrix), more draws (at least 10,000), and a fixed seed or pasted values so results are reproducible.
4. **Add Expected Shortfall** at 95% and 99%.
5. **Move to out-of-sample backtesting** with a rolling estimation window, and add Kupiec (and ideally Christoffersen) tests.
6. **Consider a fat-tailed alternative**: Student-t, EWMA/GARCH volatility, or filtered historical simulation.
7. **Clean the data**: convert text prices to numbers, check corporate-action jumps, and replace hardcoded counts with formulas.
8. **Add a short concentration note** to the dashboard, since Asian Paints and Berger drive nearly all of the risk.

10. Conclusion

The workbook is well structured and covers the standard VaR workflow end to end. Once the return-sign error is corrected, the picture changes materially: 99% VaR is higher than reported, the normal-based models look weaker, and the historical model remains the best-calibrated of the three. With the fixes above, this would be a solid, defensible market-risk project.